Tarea # 1
Yalmar Cardenas

Docker

Levantando primer contenedor

```
[cardenas@cardenas product development]$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

Docker Pull

```
[cardenas@cardenas product development]$ docker pull busybox
Using default tag: latest
latest: Pulling from library/busybox
Digest: sha256:f7ca5a32c10d51aeda3b4d01c61c6061f497893d7f6628b92f822f7117182a57
Status: Image is up to date for busybox:latest
docker.io/library/busybox:latest
```

Docker imagenes

```
[cardenas@cardenas product development]$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED       SIZE
hello-world   latest    feb5d9fea6a5   4 weeks ago   13.3kB
busybox       latest    16ea53ea7c65   6 weeks ago   1.24MB
[cardenas@cardenas product development]$
```

Docker echo

```
[cardenas@cardenas product development]$ docker run busybox echo "Hello From Gal
ileo Master"
Hello From Galileo Master
```

Docker ps

```
[cardenas@cardenas product development]$ docker ps -a
CONTAINER ID   IMAGE         COMMAND                  CREATED              STATU
S                          PORTS     NAMES
0fae7169baf6   busybox       "echo 'Hello From Ga…"   About a minute ago   Exite
d (0) About a minute ago             zealous_colden
56b0a8951062   hello-world   "/hello"                 16 minutes ago       Exite
d (0) 16 minutes ago                 zen_johnson
5abd33d982a2   hello-world   "/hello"                 18 minutes ago       Exite
d (0) 18 minutes ago                 vigilant_moore
e88a35cd2d98   hello-world   "/hello"                 20 minutes ago       Exite
d (0) 20 minutes ago                 eager_einstein
21248360a837   hello-world   "/hello"                 25 minutes ago       Exite
d (0) 25 minutes ago                 musing_leavitt
41a961885b06   hello-world   "/hello"                 26 minutes ago       Exite
d (0) 26 minutes ago                 optimistic_goldwasser
cae653901d08   hello-world   "/hello"                 13 days ago          Exite
d (0) 13 days ago                    blissful_morse
```

Docker run -it busybox

```
[cardenas@cardenas product development]$ docker run --it busybox sh
unknown flag: --it
See 'docker run --help'.
[cardenas@cardenas product development]$ docker run -it busybox sh
/ # ls
bin   dev   etc   home  proc  root  sys   tmp   usr   var
/ # uptime
 04:57:24 up  5:01,  0 users,  load average: 0.33, 0.77, 0.92
/ # exit
```

Docker pull jupyter

```
[cardenas@cardenas product development]$ docker pull jupyter/base-notebook
Using default tag: latest
latest: Pulling from jupyter/base-notebook
7b1a6ab2e44d: Pull complete
5aa2fa3625cd: Pull complete
28a7dc0afe24: Pull complete
60704649d322: Pull complete
0d122562ccc2: Pull complete
a63c0498c07b: Pull complete
ec953fac3ecc: Pull complete
6db1568369b9: Pull complete
ce14d014e8bd: Pull complete
6763b4f7efb6: Pull complete
6c1d7d331d8c: Pull complete
Digest: sha256:37f879d1621529324c890ad2fa8d8432e2ca67c62874878b9b5e93d78671a3c7
Status: Downloaded newer image for jupyter/base-notebook:latest
docker.io/jupyter/base-notebook:latest
```

Docker run

```
[cardenas@cardenas product development]$ docker run -p 8888:8888 jupyter/base-notebook
WARN: Jupyter Notebook deprecation notice https://github.com/jupyter/docker-stacks#jupyter-notebook-deprecation-notice.
Executing the command: jupyter notebook
[I 05:08:24.326 NotebookApp] Writing notebook server cookie secret to /home/jovyan/.local/share/jupyter/runtime/notebook_cookie_secr
et
[W 2021-10-27 05:08:24.943 LabApp] 'ip' has moved from NotebookApp to ServerApp. This config will be passed to ServerApp. Be sure to
 update your config before our next release.
[W 2021-10-27 05:08:24.943 LabApp] 'port' has moved from NotebookApp to ServerApp. This config will be passed to ServerApp. Be sure
to update your config before our next release.
[W 2021-10-27 05:08:24.943 LabApp] 'port' has moved from NotebookApp to ServerApp. This config will be passed to ServerApp. Be sure
to update your config before our next release.
[W 2021-10-27 05:08:24.943 LabApp] 'port' has moved from NotebookApp to ServerApp. This config will be passed to ServerApp. Be sure
to update your config before our next release.
[I 2021-10-27 05:08:24.952 LabApp] JupyterLab extension loaded from /opt/conda/lib/python3.9/site-packages/jupyterlab
[I 2021-10-27 05:08:24.952 LabApp] JupyterLab application directory is /opt/conda/share/jupyter/lab
[I 05:08:24.958 NotebookApp] Serving notebooks from local directory: /home/jovyan
[I 05:08:24.958 NotebookApp] Jupyter Notebook 6.4.5 is running at:
[I 05:08:24.958 NotebookApp] http://0c15b2fb6956:8888/?token=68a85ee0e530e84291466eb6e55d0bfb27baa84bf3626da2
[I 05:08:24.959 NotebookApp]  or http://127.0.0.1:8888/?token=68a85ee0e530e84291466eb6e55d0bfb27baa84bf3626da2
[I 05:08:24.959 NotebookApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).
[C 05:08:24.963 NotebookApp]

    To access the notebook, open this file in a browser:
        file:///home/jovyan/.local/share/jupyter/runtime/nbserver-7-open.html
    Or copy and paste one of these URLs:
        http://0c15b2fb6956:8888/?token=68a85ee0e530e84291466eb6e55d0bfb27baa84bf3626da2
     or http://127.0.0.1:8888/?token=68a85ee0e530e84291466eb6e55d0bfb27baa84bf3626da2
```

Docker ps

```
[cardenas@cardenas product development]$ docker ps
CONTAINER ID   IMAGE                   COMMAND                  CREATED         STATUS         PORTS
       NAMES
0c15b2fb6956   jupyter/base-notebook   "tini -g -- start-no…"   2 minutes ago   Up 2 minutes   0.0.0.0:8888->8888/tcp, :::8888->8888
/tcp   epic_hertz
[cardenas@cardenas product development]$
```

Jupyter lab

Docker network

```
[cardenas@cardenas product development]$ docker network create --driver bridge my_test_network
12d3b9d7538fc893d3f0f9af3e6faba02ecae537b79b0e93399f22f6e3c3a955
[cardenas@cardenas product development]$
```

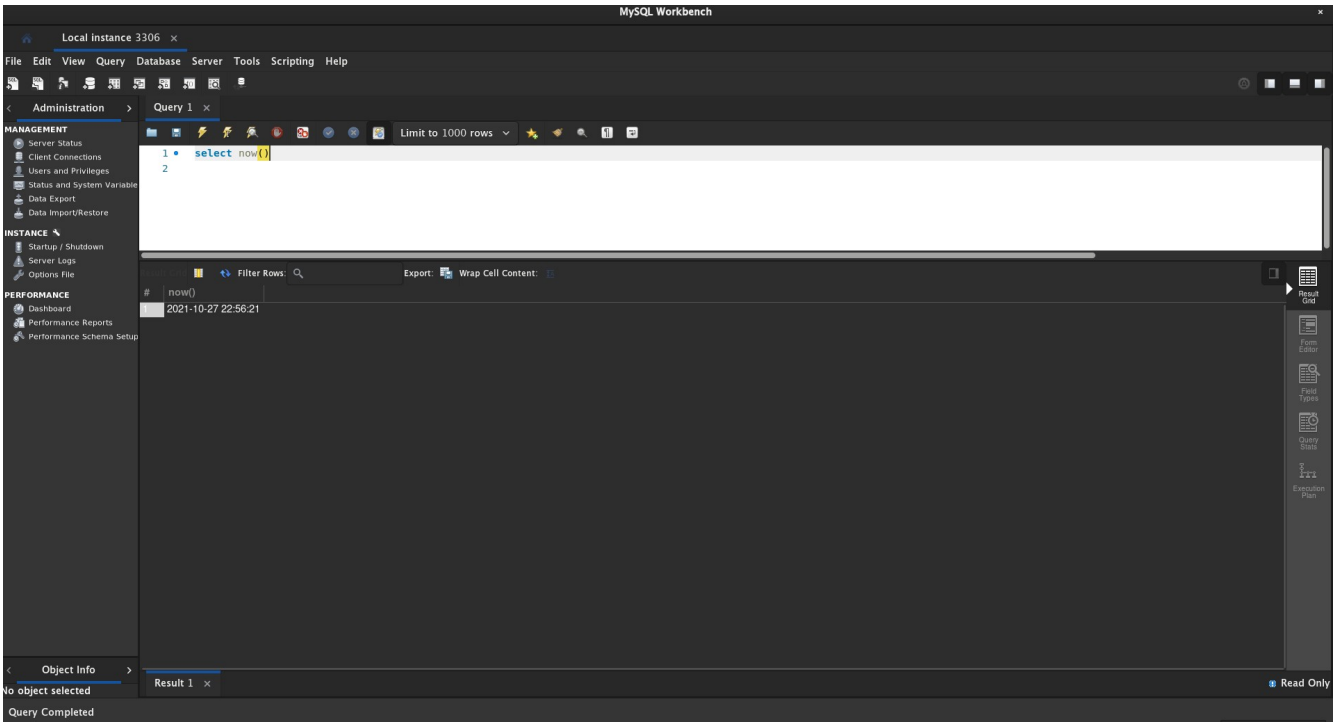
mysql

```
[cardenas@cardenas product development]$ docker run -it --network my_test_network -e "MYSQL_ROOT_PASSWORD=root123" -e "MYSQL_DATABAS
E=test" -e "MYSQL_USER=test" -e "MYSQL_PASSWORD=test123" mysql:5.7.35
Unable to find image 'mysql:5.7.35' locally
5.7.35: Pulling from library/mysql
b380bbd43752: Pull complete
f23cbf2ecc5d: Pull complete
30cfc6c29c0a: Pull complete
b38609286cbe: Pull complete
8211d9e66cd6: Pull complete
2313f9eeca4a: Pull complete
7eb487d00da0: Pull complete
bb9cc5c700e7: Pull complete
88676eb32344: Downloading [=============>                                    ]  29.03MB/108.6MB
8fea0b38a348: Download complete
3dc585bfc693: Download complete
```

mysql

```
2021-10-27T05:25:23.205941Z 0 [Note] InnoDB: Highest supported file format is Barracuda.
2021-10-27T05:25:23.226113Z 0 [Note] InnoDB: Creating shared tablespace for temporary tables
2021-10-27T05:25:23.226176Z 0 [Note] InnoDB: Setting file ./ibtmp1' size to 12 MB. Physically writing the file full; Please wait ..
.
2021-10-27T05:25:23.468017Z 0 [Note] InnoDB: File ./ibtmp1' size is now 12 MB.
2021-10-27T05:25:23.469748Z 0 [Note] InnoDB: 96 redo rollback segment(s) found. 96 redo rollback segment(s) are active.
2021-10-27T05:25:23.469784Z 0 [Note] InnoDB: 32 non-redo rollback segment(s) are active.
2021-10-27T05:25:23.470609Z 0 [Note] InnoDB: Waiting for purge to start
2021-10-27T05:25:23.520935Z 0 [Note] InnoDB: 5.7.35 started; log sequence number 12658766
2021-10-27T05:25:23.521352Z 0 [Note] InnoDB: Loading buffer pool(s) from /var/lib/mysql/ib_buffer_pool
2021-10-27T05:25:23.521934Z 0 [Note] Plugin 'FEDERATED' is disabled.
2021-10-27T05:25:23.530437Z 0 [Note] InnoDB: Buffer pool(s) load completed at 211027  5:25:23
2021-10-27T05:25:23.537832Z 0 [Note] Found ca.pem, server-cert.pem and server-key.pem in data directory. Trying to enable SSL suppor
t using them.
2021-10-27T05:25:23.537877Z 0 [Note] Skipping generation of SSL certificates as certificate files are present in data directory.
2021-10-27T05:25:23.537891Z 0 [Warning] A deprecated TLS version TLSv1 is enabled. Please use TLSv1.2 or higher.
2021-10-27T05:25:23.537899Z 0 [Warning] A deprecated TLS version TLSv1.1 is enabled. Please use TLSv1.2 or higher.
2021-10-27T05:25:23.539578Z 0 [Warning] CA certificate ca.pem is self signed.
2021-10-27T05:25:23.539651Z 0 [Note] Skipping generation of RSA key pair as key files are present in data directory.
2021-10-27T05:25:23.540697Z 0 [Note] Server hostname (bind-address): '*'; port: 3306
2021-10-27T05:25:23.540764Z 0 [Note] IPv6 is available.
2021-10-27T05:25:23.540802Z 0 [Note]   - '::' resolves to '::';
2021-10-27T05:25:23.540846Z 0 [Note] Server socket created on IP: '::'.
2021-10-27T05:25:23.549879Z 0 [Warning] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible
to all OS users. Consider choosing a different directory.
2021-10-27T05:25:23.568530Z 0 [Note] Event Scheduler: Loaded 0 events
2021-10-27T05:25:23.568923Z 0 [Note] mysqld: ready for connections.
Version: '5.7.35'  socket: '/var/run/mysqld/mysqld.sock'  port: 3306  MySQL Community Server (GPL)
```

Conexion de base de datos



jupyter y conexion de red

```
[W 2021-10-27 23:00:01.047 LabApp] 'ip' has moved from NotebookApp to ServerApp. This config will be passed to ServerApp.
re to update your config before our next release.
[W 2021-10-27 23:00:01.047 LabApp] 'port' has moved from NotebookApp to ServerApp. This config will be passed to ServerApp
sure to update your config before our next release.
[W 2021-10-27 23:00:01.047 LabApp] 'port' has moved from NotebookApp to ServerApp. This config will be passed to ServerApp
sure to update your config before our next release.
[W 2021-10-27 23:00:01.047 LabApp] 'port' has moved from NotebookApp to ServerApp. This config will be passed to ServerApp
sure to update your config before our next release.
[I 2021-10-27 23:00:01.055 LabApp] JupyterLab extension loaded from /opt/conda/lib/python3.9/site-packages/jupyterlab
[I 2021-10-27 23:00:01.055 LabApp] JupyterLab application directory is /opt/conda/share/jupyter/lab
[I 23:00:01.060 NotebookApp] Serving notebooks from local directory: /home/jovyan
[I 23:00:01.060 NotebookApp] Jupyter Notebook 6.4.5 is running at:
[I 23:00:01.060 NotebookApp] http://160adbf505f3:8888/?token=14bfca64299c0eaf989494ee8a31498c52d5063df427f375
[I 23:00:01.060 NotebookApp]  or http://127.0.0.1:8888/?token=14bfca64299c0eaf989494ee8a31498c52d5063df427f375
[I 23:00:01.060 NotebookApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).
[C 23:00:01.064 NotebookApp]

    To access the notebook, open this file in a browser:
        file:///home/jovyan/.local/share/jupyter/runtime/nbserver-7-open.html
    Or copy and paste one of these URLs:
        http://160adbf505f3:8888/?token=14bfca64299c0eaf989494ee8a31498c52d5063df427f375
     or http://127.0.0.1:8888/?token=14bfca64299c0eaf989494ee8a31498c52d5063df427f375
[I 23:00:26.657 NotebookApp] 302 GET /?token=14bfca64299c0eaf989494ee8a31498c52d5063df427f375 (172.18.0.1) 0.860000ms
```

Entrar a contenedor

```
[cardenas@cardenas ~]$ docker exec -it recursing_heisenberg sh
$
```

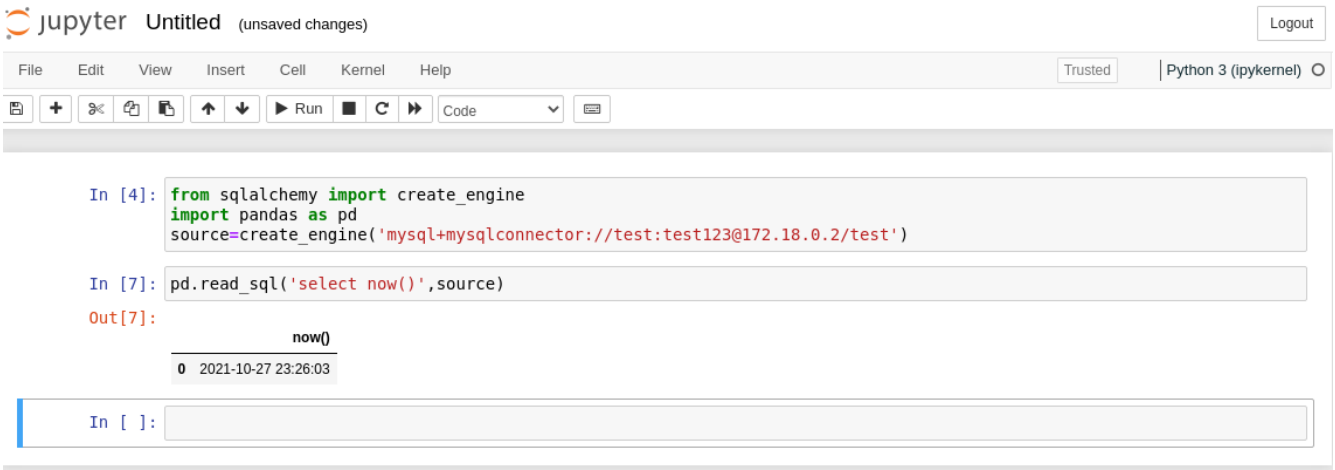
Instalando conector

```
$ pip install mysql-connector-python
Collecting mysql-connector-python
  Downloading mysql_connector_python-8.0.27-1commercial-cp39-cp39-manylinux1_x86_64.whl (37.5 MB)
     |[redacted]| 37.5 MB 104 kB/s
Collecting protobuf>=3.0.0
  Downloading protobuf-3.19.0-cp39-cp39-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (1.1 MB)
     |[redacted]| 1.1 MB 1.3 MB/s
Installing collected packages: protobuf, mysql-connector-python
Successfully installed mysql-connector-python-8.0.27 protobuf-3.19.0
$
```

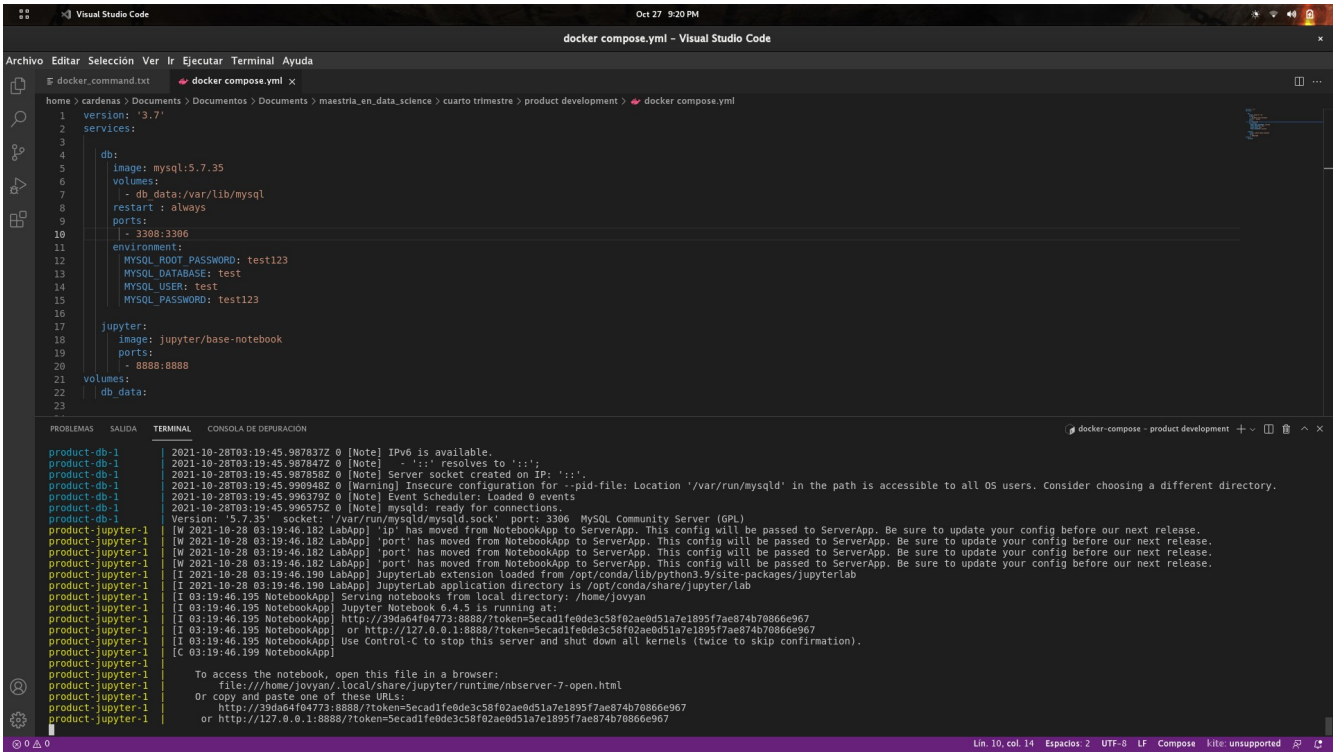
Instalando pandas

```
$ pip install pandas
Collecting pandas
  Downloading pandas-1.3.4-cp39-cp39-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (11.5 MB)
     |[redacted]| 11.5 MB 192 kB/s
Requirement already satisfied: pytz>=2017.3 in /opt/conda/lib/python3.9/site-packages (from pandas) (2021.3)
Collecting numpy>=1.17.3
  Downloading numpy-1.21.3-cp39-cp39-manylinux_2_12_x86_64.manylinux2010_x86_64.whl (15.7 MB)
     |[redacted]| 10.9 MB 328 kB/s eta 0:00:15
```

Conexion a mysql



Docker compose



coneccion a base de datos desde jupyter

127.0.0.1:8888/notebooks/work/Untitled.ipynb?kernel_name=python3

jupyter Untitled (unsaved changes)

Logout

File Edit View Insert Cell Kernel Help Trusted Python 3 (ipykernel)

Run Code

In [1]:
```
!pip install mysql-connector-python
```

```
Collecting mysql-connector-python
  Downloading mysql_connector_python-8.0.27-1commercial-cp39-cp39-manylinux1_x86_64.whl (37.5 MB)
     |████████████████████████████████| 37.5 MB 190 kB/s
Collecting protobuf>=3.0.0
  Downloading protobuf-3.19.0-cp39-cp39-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (1.1 MB)
     |████████████████████████████████| 1.1 MB 250 kB/s
Installing collected packages: protobuf, mysql-connector-python
Successfully installed mysql-connector-python-8.0.27 protobuf-3.19.0
```

In [2]:
```
!pip install pandas
```

```
Collecting pandas
  Downloading pandas-1.3.4-cp39-cp39-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (11.5 MB)
     |████████████████████████████████| 11.5 MB 96 kB/s
Collecting numpy>=1.17.3
  Downloading numpy-1.21.3-cp39-cp39-manylinux_2_12_x86_64.manylinux2010_x86_64.whl (15.7 MB)
     |████████████████████████████████| 15.7 MB 129 kB/s
Requirement already satisfied: python-dateutil>=2.7.3 in /opt/conda/lib/python3.9/site-packages (from pandas) (2.
8.2)
Requirement already satisfied: pytz>=2017.3 in /opt/conda/lib/python3.9/site-packages (from pandas) (2021.3)
Requirement already satisfied: six>=1.5 in /opt/conda/lib/python3.9/site-packages (from python-dateutil>=2.7.3->pa
ndas) (1.16.0)
Installing collected packages: numpy, pandas
Successfully installed numpy-1.21.3 pandas-1.3.4
```

In [2]:
```
from sqlalchemy import create_engine
import pandas as pd
```

In [4]:
```
source = create_engine('mysql+mysqlconnector://test:test123@db/test')
```

In [8]:
```
pd.read_sql('select now()',source)
```

Out[8]:

	now()
0	2021-10-28 03:54:20

In [ ]: